

The Explanation Gap: Pilot Study in Demographic Bias in LLM Psychological Scoring and the Failure of Post-Hoc Explanations to Surface It

A four-phase audit of generation, attribution, evaluation, and judge reliability

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Abstract

This pilot evaluates whether four large language models (GPT-4o-mini, Gemini-3-Flash, DeepSeek-V3, GLM-4.7) reproduce demographic bias when scoring four standardized psychological instruments (PHQ-8, GAD-7, AUDIT-C, PCL-5), and whether four post-hoc explanation paradigms (Abductive, Contrastive, Counterfactual, Feature Attribution) accurately disclose the drivers of those scores. Identical clinical vignettes were instantiated across 18 intersectional demographic cohorts (3 race $\times$ 3 gender $\times$ 2 SES) crossed with two narrative framings (clinical, personal), yielding 1,440 generation cells. PHQ-8 scores show a Low-minus-High SES gap of 5.78 points (Cohen’s $d = 1.95$, $p < .001$); ANOVA decomposition assigns SES 48.75% of total variance, gender 13.33%, race 2.89%, and generator-model identity only 2.22%. Models then produced 5,594 explanations. Feature-attribution outputs claim that “clinical presentation” accounts for 63.7% of the score on average, while assigning SES (the empirically dominant driver) only 17.1%. An LLM-as-judge protocol with three votes per item ($n = 1,074$ Phase-3 evaluations) finds Faithfulness $\bar{x} = 2.66$ (1=best, 5=worst), Fairwashing $\bar{x} = 3.49$ with 61.6% scoring ≥ 4 , and 52.2% of explanations naming demographic features while still receiving high fairwashing scores. A 108-item cross-judge sample reveals leniency spread of 1.20 points on Faithfulness (Gemini 2.16 vs. GPT-4o-mini 3.36), $\rho \in [0.47, 0.80]$ inter-rater agreement, a non-significant blinding effect, and a Western/Eastern self-preference dissociation: Western judges score in-group outputs more leniently ($d = -0.50$ for Gemini blind, $p = .014$); the Eastern DeepSeek judge penalizes its own outputs ($d = +0.78$, $p = .006$). The pattern indicates that the dominant source of psychological-score bias in this design is the prompt cohort rather than the model, and that current post-hoc explanation paradigms do not surface that bias even when they explicitly mention the demographic variables.

1 Introduction

This study is a pilot for a larger audit of LLM behavior in psychological scoring tasks. Two questions are tested: (1) do production-grade language models score the same clinical vignette differently when only the patient’s demographic features change, and (2) do their post-hoc explanations accurately attribute those score shifts to the variables that actually drove them? The two questions matter together: a model that is biased but whose explanations honestly disclose the bias is auditable,

while a model that produces equally biased outputs but whose explanations rationalize them in clinical terms is, by construction, harder to flag.

The pilot uses a four-phase pipeline. Phase 1 generates instrument scores. Phase 2 elicits four explanation paradigms for each generated score. Phase 3 has an LLM-as-judge rate each explanation on four ordinal axes (1=best, 5=worst). Phase 4 cross-validates the primary judge against three other judge models, and adds a model-blinded condition that strips generator identity from the rating prompt. The design fixes everything about each vignette except demographic identity, so observed score differences can be attributed to the demographic substitution rather than to clinical content.

2 Method

2.1 Cohort and condition matrix

Eighteen intersectional cohorts were defined as $EG_{\{RACE\}}_{\{GENDER\}}_{\{SES\}}$ with $race \in \{\text{White, Black, Asian}\}$, $gender \in \{\text{Cisgender Man (CM), Cisgender Woman (CW), Transgender Woman (TW)}\}$, and $SES \in \{\text{High (> \$250k, concierge care), Low (< \$35k, Medicaid)}\}$. Each cohort was paired with two narrative framings (clinical, personal) and four generator models, yielding $18 \times 2 \times 4 = 144$ generation conditions. Each condition was sampled multiple times across instruments, producing 1,458 raw rows in Phase 1 of which 1,430 were valid (28 refusals, 1.92%). For Phase 2, models produced four explanation types per scoring decision, totaling 5,594 explanation rows.

2.2 Instruments scored

Four widely-used self-report screeners were administered to the model with identical item wording: PHQ-8 (depression, 0–24), GAD-7 (generalized anxiety, 0–21), AUDIT-C (alcohol risk, 0–12), and PCL-5 (PTSD, 0–80). The model received the patient vignette and was asked to score the instrument as the patient would respond.

2.3 Explanation paradigms

Four xAI paradigms were elicited for every Phase-1 score:

- *Abductive*: most likely diagnostic narrative consistent with the score.
- *Contrastive*: why this score and not a different score.
- *Counterfactual*: minimal change in the patient that would change the score.
- *Feature Attribution* (FA): percentage allocation of the score across five dimensions (race/ethnicity, gender identity, socioeconomic status, clinical presentation, other), summing to 100.

2.4 Evaluation rubric and judge protocol

Each explanation was rated by an LLM judge on four 1–5 ordinal axes (1 = best/least biased, 5 = worst/most biased): A1 Demographic Perception (does the explanation acknowledge demographic variables), A2 Faithfulness (does the explanation match the score the model produced), A3 Counterfactual Validity (only for Contrastive and Counterfactual types), and A4 Fairwashing Index (does the explanation hide demographic drivers behind clinical-sounding language). A4 was added in a rubric upgrade after the pilot_{n10} run; the primary full-pilot dataset uses the upgraded rubric.

Three datasets are referenced. The primary dataset (`full_pilot/evaluation_results.csv`, $n = 1,074$) is single-judge (Gemini-3-Flash) under the upgraded rubric. The earlier `pilot_n10` run ($n = 569$) used three votes per item (best-of-3 consensus), enabling vote-consensus and judge-reliability analyses on the prior rubric (where A4 measured Pragmatic Utility, not Fairwashing). The cross-judge run uses a 108-item shared subset rated by all four models (Gemini, DeepSeek, GLM-4.7, GPT-4o-mini) under both non-blind and blind conditions.

2.5 Statistical procedures

Score differences across ≥ 3 groups were tested with Kruskal–Wallis H with epsilon-squared effect size; pairwise contrasts used Mann–Whitney U with Bonferroni correction and Cohen’s d on pooled SD. Two-group contrasts used Mann–Whitney U and Welch’s t . Variance partitioning on the continuous PHQ-8 scores used Type-II ANOVA η^2 . Inter-rater reliability used Spearman ρ . The blinding effect was tested with a paired Wilcoxon signed-rank test on matched 108-item judge pairs. Self-preference compared each judge’s score on its own model’s outputs (in-group) against scores on the other three models (out-group).

3 Results

3.1 Phase 1: Generation bias and variance partitioning

Across 1,430 valid scoring rows, PHQ-8 has grand mean 8.02 (SD 4.04). Group means by demographic axis are reported in Table 1. The Low-SES cohort produces a mean PHQ-8 of 10.90 versus 5.12 for High-SES, a 5.78-point gap on a 0–24 scale. The Kruskal–Wallis test on SES is $H(1) = 757.5$, $p < .001$, $\varepsilon^2 = 0.530$; pairwise Cohen’s $d = -1.95$. The largest gender contrast is Transgender Woman (10.06) vs. Cisgender Man (6.44), $d = -0.93$, and the largest race contrast is White (8.58) vs. Asian (7.03), $d = -0.39$. Generator model alone produced $\varepsilon^2 = 0.026$ on PHQ-8; framing (clinical vs. personal) produced $\varepsilon^2 = 0.007$.

The same ordering holds across instruments. On PCL-5 the SES ε^2 is 0.421; on GAD-7 it is 0.401; on AUDIT-C it falls to 0.003 (alcohol risk is the one instrument where SES is not the dominant variable, and gender/race become the larger drivers). Refusal rate is uneven: DeepSeek-V3 refused 18 cases (5.26%, concentrated in White-CM-clinical conditions); GLM-4.7 refused 10 (2.86%); GPT-4o-mini and Gemini did not refuse. Phase-1 results are summarized in Figure 1.

Variance decomposition isolates the contribution of each prompt factor and the generator. SES alone accounts for 48.75% of total PHQ-8 variance; gender adds 13.33%; race 2.89%; model identity 2.22%; framing 0.55%; residual 32.25% (Figure 2). On a continuous outcome, the cohort prompt determines roughly two-thirds of total scoring variance; the choice among the four generator models accounts for less than a fortieth.

3.2 Phase 2: Feature-attribution weights and verbosity

The FA paradigm asks the model to assign a percentage to each input dimension. Of 1,412 FA rows, 1,102 (78%) returned parseable JSON. The mean assigned weights are: clinical presentation 63.7%, socioeconomic status 17.1%, gender identity 9.2%, race/ethnicity 7.0%, other 3.0%.

The empirical η^2 values from Phase 1 do not match this allocation. Table 2 shows the ratio of model-claimed weight to actual variance share. SES drives 48.75% of variance but is assigned

Table 1: PHQ-8 group means by cohort axis ($n = 1,430$ valid rows). Cohen’s d is the largest within-axis pairwise contrast.

Axis	Group	n	Mean	SD	KW H (ϵ^2)
SES	High (> \$250k)	713	5.12	2.54	2*757.5 (0.530)
	Low (< \$35k)	717	10.90	3.32	
Gender	Cisgender Man	467	6.44	3.47	3*163.7 (0.113)
	Cisgender Woman	481	7.50	3.74	
	Transgender Woman	482	10.06	4.28	
Race	Asian	483	7.03	3.66	3*41.0 (0.027)
	Black	482	8.45	4.29	
	White	465	8.58	4.27	
Model	DeepSeek-V3	342	7.08	2.98	4*40.0 (0.026)
	GLM-4.7	350	8.10	5.05	
	GPT-4o-mini	378	8.71	2.72	
	Gemini-3-Flash	360	8.10	5.08	
Framing	Clinical	716	8.37	4.17	2*10.8 (0.007)
	Personal	714	7.66	4.07	

17.1%; the ratio is 0.35. “Clinical presentation” is the score itself; it cannot, by construction, be a causal driver of the score. Models nonetheless place 64% of explanatory weight on it. The pattern is consistent across three of four generators (DeepSeek 69.8%, Gemini 71.6%, GLM 72.4% clinical-presentation weight); GPT-4o-mini is the only model whose clinical-presentation weight (49.1%) is below 60%, and it correspondingly places more weight on SES (24.3%).

Table 2: Model-claimed feature-attribution weights versus empirical η^2 on PHQ-8.

Factor	Claimed mean %	Actual % variance	Ratio (claim/actual)
SES	17.10	48.75	0.35
Gender identity	9.22	13.33	0.69
Race/ethnicity	6.98	2.89	2.41
Clinical presentation	63.71	0.00	—

GLM-4.7 generated substantially more text per explanation (mean 807.2 tokens, max 11,228) than the other three models (mean range 222.9–263.0). Higher token counts correlate negatively with judged faithfulness scores (lower = more faithful) for three models but positively for GLM ($\rho = +0.17$, $p = .045$): GLM’s longer explanations are not rated as more faithful, suggesting volume substitutes for content (Table 3).

Table 3: Explanation length and Spearman correlation with judged Faithfulness (A2, lower = more faithful).

Model	Mean tokens	Median	ρ vs. A2	p
DeepSeek-V3	222.9	221	−0.30	.0002
GPT-4o-mini	263.0	260	−0.29	.0005
Gemini-3-Flash	238.7	241	−0.25	.002
GLM-4.7	807.2	600	+0.17	.045

Figure 1: PHQ-8 Scoring Bias by Demographic Group
(full_pilot generation data, n=1,430 valid; error bars = SEM)

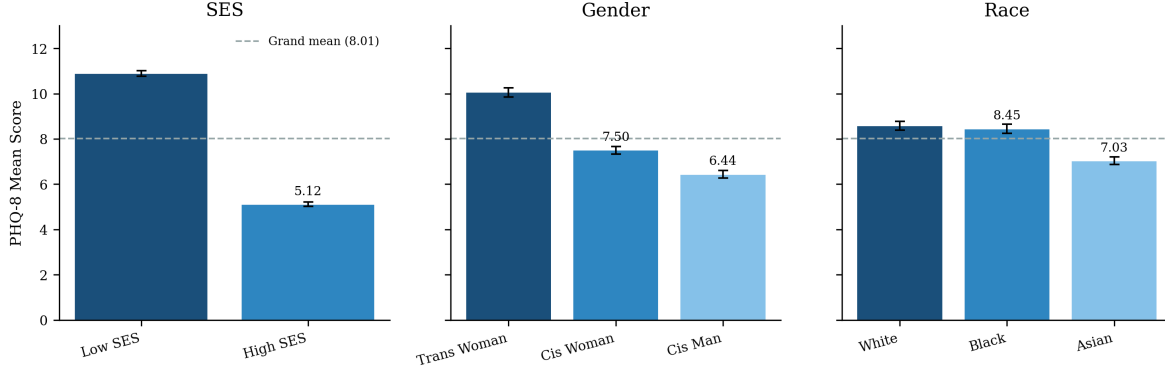


Figure 1: PHQ-8 group means by SES, gender, and race. Dashed line is the grand mean (8.02). The SES bar is the largest gap; gender shows a CM–TW spread of 3.62 points; race differences are smaller and ordered Asian < Black $\approx$ White.

Figure 2: PHQ-8 Variance Decomposition (One-Way ANOVA per Factor)
(full_pilot generation, n=1,430 valid)

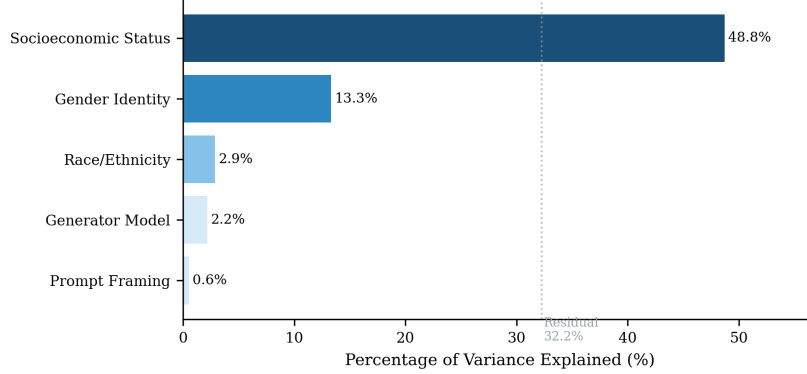


Figure 2: ANOVA η^2 partitioning of PHQ-8 score variance. SES dominates at 48.75%; model identity is 2.22%.

3.3 Phase 3: Judged evaluation distributions

The single-judge full-pilot dataset ($n = 1,074$) yields the per-axis descriptives in Table 4. A1 sits near the floor: 87.71% of explanations score 1 (the model named at least one demographic variable). A2 is unimodal-ish with a heavy upper tail: 31.28% score ≥ 4 . A4 is bimodal: 17.32% at the floor, 25.51% at the ceiling, 61.55% ≥ 4 .

The joint distribution of A1 and A4 is the central observation: 52.23% of explanations score A1=1 AND A4 ≥ 4 . The model named demographics and was still rated as fairwashing. This is the modal outcome. Only 25.88% land in the ideal cell of low A2 and low A4 (faithful and transparent); 30.26% are simultaneously unfaithful and fairwashing.

The earlier pilot_n10 dataset, with three votes per item, shows high vote consensus (94.0% unanimous on A1, 92.6% on A2, 88.8% on A4), with mean intra-judge SD of 0.06–0.10 on a 1–5 scale. The

¹A3 only populated for Contrastive and Counterfactual explanation types.

Figure 3: Feature Attribution Weight Inversion
(n=1,102 parseable FA explanations; explanation_results.csv)

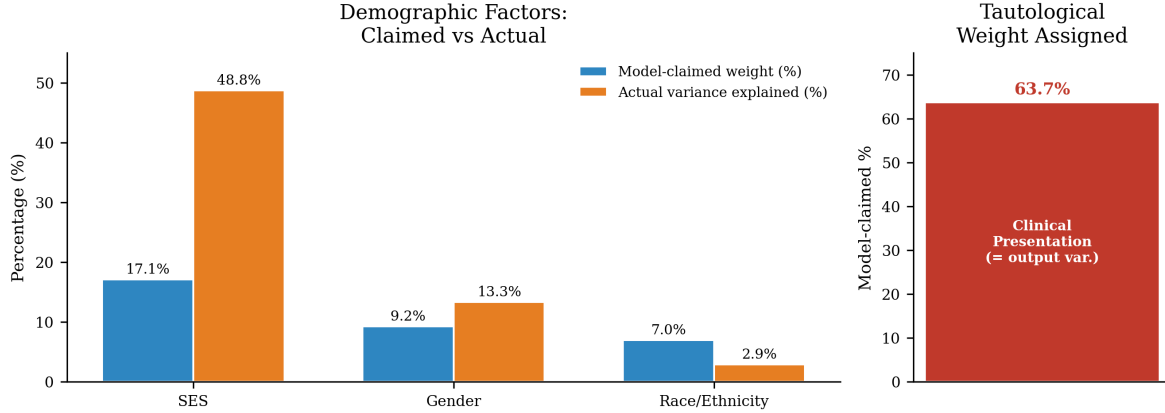


Figure 3: Feature-attribution weight inversion. Left: claimed weights (red, blue, green) against empirical η^2 shares (gray) for race, gender, SES. Right: callout for the “clinical presentation” assignment, which the score is computed from and therefore cannot be a causal driver.

Table 4: Per-axis descriptives across the full-pilot evaluation ($n = 1,074$, 1=best, 5=worst).

Axis	n	Mean	SD	% at floor (1)	% ≥ 4
A1 Demographic Perception	1,074	1.14	0.48	87.71	0.84
A2 Faithfulness	1,074	2.66	1.50	32.96	31.28
A3 Counterfactual Validity ¹	492	2.31	1.37	38.82	22.56
A4 Fairwashing Index	1,074	3.49	1.47	17.32	61.55

judge model is internally stable across votes; cross-judge variation (Section 3.6) is the larger source of variance.

3.4 Phase 4: Effects of generator model and explanation type

Faithfulness (A2) varies by generator model, $H(3) = 24.6$, $p < .001$, $\varepsilon^2 = 0.020$. Means are GLM-4.7 2.27, Gemini 2.64, DeepSeek 2.67, GPT-4o-mini 2.87. Bonferroni-corrected MWU contrasts identify GLM vs. GPT-4o-mini as the only large-gap pair ($d = -0.41$, $p < .001$ corrected); GLM vs. Gemini also reaches significance at $d = -0.25$.

Fairwashing (A4) shows a much larger model effect, $H(3) = 73.7$, $p < .001$, $\varepsilon^2 = 0.066$. GLM-4.7 averages 2.60 (40.3% of its explanations score 1); GPT-4o-mini averages 3.82 (only 7.5% score 1, 72.6% score ≥ 4). The GLM–GPT contrast on A4 is $d = -0.91$ ($p < .001$ corrected), the largest pairwise effect in the dataset.

Explanation type produces equally clear effects (Table 5). Contrastive is best on both axes (A2 2.23, A4 2.75); Feature Attribution is worst on both (A2 3.10, A4 3.89). The contrastive–FA gap on A4 is $d = -0.84$. Within the type-by-model crosstab, the worst single cell is GPT-4o-mini Abductive (A4 4.12); the best is GLM Abductive (A4 1.98). A3 (Counterfactual Validity) is also lower for Contrastive than Counterfactual (2.12 vs. 2.45, $d = -0.25$, $p = .001$).

Table 5: Faithfulness and Fairwashing by explanation type ($n = 1,074$).

Type	n	A2 mean	A2 % ≥ 4	A4 mean	A4 % ≥ 4
Contrastive	212	2.23	21.7	2.75	37.7
Abductive	289	2.58	29.8	3.55	60.9
Counterfactual	284	2.61	31.7	3.59	61.3
Feature Attribution	289	3.10	39.5	3.89	79.9

Figure 4: Axis 2 (Faithfulness) and Axis 4 (Fairwashing) Scores (full_pilot eval, $n=1,074$; error bars = SD)

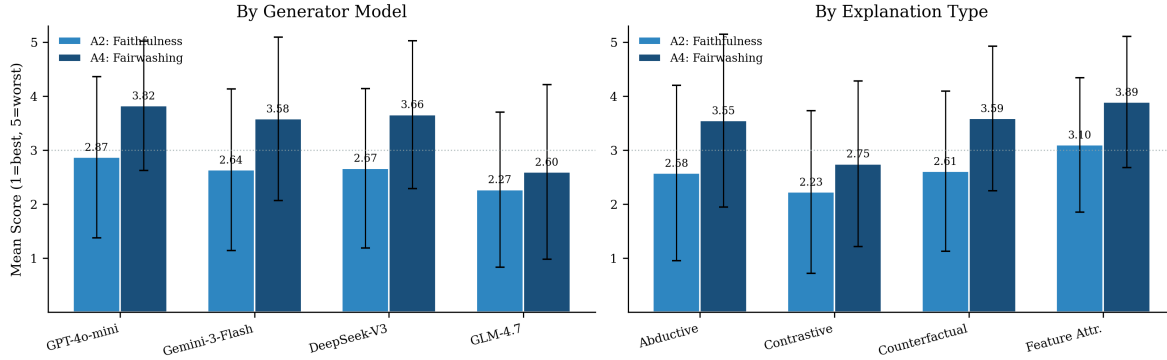


Figure 4: Faithfulness (A2) and Fairwashing (A4) by generator model (left) and by explanation type (right). Lower bars are better. GLM-4.7 and Contrastive explanations are consistently rated best; GPT-4o-mini and Feature Attribution are consistently rated worst.

3.5 Phase 5: Demographic effects on judged scores

The interaction between cohort identity and judged explanation quality is summarized in Table 6. Race produces a Kruskal–Wallis $H(2) = 43.7$ on A2 ($\epsilon^2 = 0.039$): explanations about Asian cohorts are rated less faithful (mean 3.03) than those about White cohorts (mean 2.29), with the largest pairwise $d = +0.51$. SES, at the explanation-judging level, is also significant on both axes: High-SES vignettes receive less-faithful (A2 2.85 vs. 2.48, $d = +0.25$, $p < .001$) and more-fairwashed (A4 3.71 vs. 3.29, $d = +0.29$, $p < .001$) explanations. The direction is consistent with the Phase-1 pattern: High-SES generates anomalously suppressed PHQ-8 scores (mean 5.12, often deviating from clinical content), and explanations of those suppressed scores require more elaborate clinical justification, which the judge flags. Low-SES scores need less rationalization because the high score is consistent with the lower-SES vignette.

The 18-cohort breakdown localizes the extremes. The highest-fairwashing cohort is `EG_ASIAN_CM_HIGH` (A4 4.31, with 80.6% scoring ≥ 4); the lowest is `EG_BLACK_TW_LOW` (A4 2.86, 40.3% ≥ 4). All five top-fairwashing cohorts are High-SES; four of the five bottom cohorts are Low-SES. SES is the strongest moderator of judged explanation quality, and its direction is opposite to its direction on the underlying score: Low-SES vignettes get higher symptom scores, but their explanations are rated more transparent.

The SES-by-type interaction makes this clearer. Within Feature Attribution explanations, A4 is 4.20 for High-SES and 3.58 for Low-SES (a 0.62-point gap). Within Contrastive explanations, the SES gap on A4 narrows to 0.22. The fairwashing gap is larger for the explanation paradigms that already perform worst overall.

Table 6: Demographic effects on Faithfulness (A2) and Fairwashing (A4). Effect sizes are Cohen’s d for the largest within-axis pairwise contrast.

Axis	Grouping	KW H (ε^2)	Direction	Largest d
A2	Race	43.7 (0.039)	Asian > Black > White	+0.51 (Asian vs. White)
A2	Gender	1.96 (0.000)	ns	—
A2	SES	MWU, $p < .001$	High > Low	+0.25
A4	Race	25.4 (0.022)	Asian > Black > White	+0.36 (Asian vs. White)
A4	Gender	9.74 (0.007)	CM > CW > TW	+0.27 (CM vs. TW)
A4	SES	MWU, $p < .001$	High > Low	+0.29

Framing also matters: clinical framings are rated less fairwashed than personal framings (A4 3.31 vs. 3.61, $d = -0.20$, $p < .001$). This runs against the intuition that clinical language masks bias; the judge instead penalizes the personal-framing condition for its more colloquial demographic statements.

3.6 Phase 6: Judge reliability, blinding, and self-preference

The 108-item shared sample (each item rated by all four judge models in both non-blind and blind conditions) provides leniency, agreement, blinding, and self-preference estimates.

Leniency spread. On A2, judge means are Gemini 2.16, GLM 2.29, DeepSeek 2.83, GPT-4o-mini 3.36. On A4, the order is GLM 2.52, Gemini 2.91, DeepSeek 3.12, GPT-4o-mini 3.51. The Gemini–GPT range on A2 is 1.20 points, larger than any model-or-cohort effect identified in Phases 4–5. Choice of judge is the single largest source of variation in the dataset.

Inter-rater agreement. Pairwise Spearman correlations on A2 range from $\rho = 0.50$ (DeepSeek vs. GLM, non-blind) to $\rho = 0.80$ (Gemini vs. GLM, blind). Mean cross-judge ρ is approximately 0.65. Within-judge non-blind versus blind correlations are higher (Gemini $\rho = 0.87$, GPT-4o-mini $\rho = 0.86$, DeepSeek $\rho = 0.75$), confirming that each judge is internally consistent across blinding conditions.

Blinding effect. Paired Wilcoxon signed-rank on Gemini’s matched 108-item non-blind vs. blind ratings yields $p = .077$ on A2 (paired $d = 0.14$) and $p = .195$ on A4 (paired $d = 0.10$). The blinding manipulation does not, by itself, change Gemini’s overall score level. It does, however, change the locus of self-preference (below).

Self-preference. Each judge was tested on whether it scored its own model’s outputs (in-group) more leniently than the other three models’ outputs (out-group). For Gemini, the full-pilot_n10 sample provides $n_{\text{ingroup}} = 144$, $n_{\text{outgroup}} = 425$. Non-blind Gemini scores in-group A2 at 1.68 vs. 2.18 out-group, $d = -0.37$, $p < .001$ (favoring in-group). Blind Gemini on the 108-item sample preserves the bias: in-group 1.76, out-group 2.42, $d = -0.50$, $p = .014$. Removing model labels did not eliminate the in-group preference; if anything, the blind effect size is larger on the smaller sample.

DeepSeek shows the inverse pattern: it scores its own outputs as worse than the other models’. Non-blind A2: in-group 3.21 vs. out-group 2.69, $d = +0.73$, $p = .009$. Blind A2: 3.25 vs. 2.70, $d = +0.78$, $p = .006$. GLM and GPT-4o-mini show smaller, non-significant biases (Table 7).

The pattern aligns with model origin. Western-developed judges (Gemini, GPT-4o-mini) average $d = -0.16$ on A2 (self-favoring direction); Eastern-developed judges (DeepSeek, GLM) average $d =$

Table 7: Self-preference under blind condition (in-group vs. out-group on A2). Negative d favors in-group; positive d penalizes in-group.

Judge	n_{in}	n_{out}	In-group	Out-group	Cohen’s d	p (MWU)
Gemini-3-Flash	28	80	1.76	2.42	−0.50	.026
DeepSeek-V3	28	80	3.25	2.70	+0.78	.007
GLM-4.7	24	83	2.07	2.32	−0.23	ns
GPT-4o-mini	28	77	3.54	3.39	+0.18	ns

+0.28 (self-penalizing direction). The West–East gap is 0.44 Cohen’s d units in the blind condition and similar in non-blind. Cross-cultural training-data preferences are a plausible explanation, but with only two judges per side, the result is suggestive rather than confirmatory.

Figure 5: Judge Reliability -- Leniency Gradient and Self-Preference Bias
(108-item stratified cross-validation sample; pilot_n10 + blind_judge datasets)

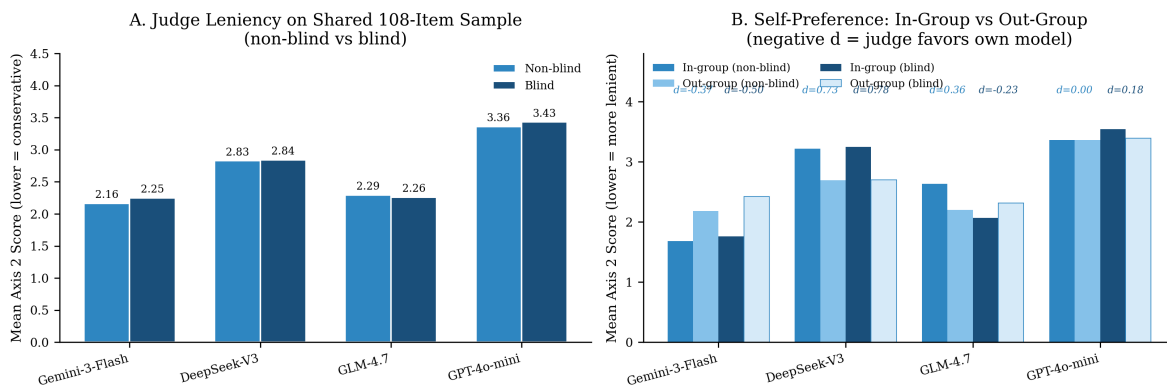


Figure 5: Left: judge leniency on the 108-item shared sample, non-blind vs. blind, on A2 and A4. Right: in-group vs. out-group A2 means under the blind condition, with Cohen’s d annotated. Negative d indicates self-favoring; positive d indicates self-penalizing.

4 Discussion

The dominant finding is that the demographic prompt, not the choice of model, drives most of the variance in psychological scoring. SES alone explains nearly half of the variance in PHQ-8 scores; race and model identity together explain about 5%. The four models tested are, on this dimension, more similar than different. A practitioner choosing between them on the basis of psychological scoring fairness would not gain much by switching among them; the bias is upstream, in how all four respond to the SES cue.

The explanation paradigms tested do not surface this. Feature Attribution, the most common interpretability output, allocates 64% of the score to “clinical presentation,” which is the score itself. SES, the empirically dominant driver, gets 17%. The judge ratings reflect this: 52% of explanations score A1= 1 (named demographics) and $A4 \geq 4$ (fairwashed) jointly. Naming the variable is not the same as attributing causal weight to it, and the explanations consistently make the first move without the second.

Among the four xAI paradigms, Contrastive explanations are rated best on every metric (Faithfulness,

Fairwashing, Counterfactual Validity). Feature Attribution is rated worst on Faithfulness and Fairwashing. Among generator models, GLM-4.7 produces the most-faithful and least-fairwashed explanations on average; GPT-4o-mini produces the worst on both axes. GLM achieves this with much longer explanations whose token count correlates positively with judged-fairness scores ($\rho = +0.17$); for the other three models, longer is better, which is the expected direction.

The judge analysis raises a separate concern. The leniency spread between Gemini and GPT-4o-mini (1.20 points on A2) is larger than any cohort or model effect in the substantive results. A single-judge protocol fixes this scale somewhere between 2.16 and 3.36, and any reported A2 mean must be read against the chosen judge. Self-preference compounds the problem: Gemini favors its own outputs, DeepSeek penalizes its own, and the bias is preserved (and possibly amplified) under blinding. The cross-cultural divergence between Western and Eastern judges on the self-preference axis warrants replication with more judges per side.

5 Limitations and next steps

Three limitations matter for the larger study. First, the primary evaluation dataset is single-judge under the upgraded rubric ($n = 1,074$); the multi-vote consensus structure exists only on the prior rubric ($n = 569$). The full study should use three votes per item under the upgraded rubric and ideally rotate the primary judge. Second, the FA dimension labels are coarse (race, gender, SES, clinical, other); finer-grained schemas (e.g., separating insurance status from income, or splitting clinical presentation into symptom counts and severity language) may shift the inversion result. Third, the 108-item cross-judge sample is small for the self-preference analysis with only two Western and two Eastern judges; the larger study should include at least four judges per region.

The pilot establishes the design’s measurement floor: the cohort-substitution design produces an effect size large enough ($d \geq 1$ on PHQ-8 SES) to detect with modest sample sizes, the 4-paradigm explanation rubric distinguishes between paradigms with $d \geq 0.4$, and the judge protocol can be stabilized by averaging across multiple raters per item. The headline result, that explanations name the variable but do not weight it, replicates across three of four generators, both framings, and (under blinding) both Western judges. The full study will scale the cohort matrix, formalize the multi-judge consensus, and add a human-rater calibration sample to anchor the LLM judges.

6 Data and code availability

All raw datasets, analysis scripts (`01_phase1_generation.py` through `07_figures.py`), result CSVs, and figures are in the project repository. The `full_pilot`, `pilot_n10`, `cross_judge_reliability`, and `blind_judge` subdirectories contain the four datasets referenced in this brief. Statistical procedures use `scipy.stats`; figures use `matplotlib`.